

Name:P.Tejaswini
Rollno:2520030011

Order table_sql aggregate and clause operations:

Program:

```
1      -- Step 1: Create Database and Table
2  ●    CREATE DATABASE orders_management_db;
3  ●    USE orders_management_db;
4  ●    CREATE TABLE orders (
5      ord_no INT PRIMARY KEY,
6      purch_amt DECIMAL(10,2),
7      ord_date DATE,
8      customer_id INT,
9      salesman_id INT
10     );
11     -- Step 2: Insert Data
12  ●    INSERT INTO orders VALUES
13      (70001,150.50,'2012-10-05',3005,5002),
14      (70009,270.65,'2012-09-10',3001,5005),
15      (70002,65.26,'2012-10-05',3002,5001),
16      (70004,110.50,'2012-08-17',3009,5003),
17      (70007,948.50,'2012-09-10',3005,5002),
18      (70005,2400.60,'2012-07-27',3007,5001),
19      (70008,5760.00,'2012-09-10',3002,5001),
20      (70010,1983.43,'2012-10-10',3004,5006),
21      (70003,2480.40,'2012-10-10',3009,5003),
22      (70012,250.45,'2012-06-27',3008,5002),
23      (70011,75.29,'2012-08-17',3003,5007),
24      (70013,3045.60,'2012-04-25',3002,5001);
```

Output:

Output				
Action Output				
#	Time	Action	Message	Duration / Fetch
✓ 1	19:19:37	CREATE DATABASE orders_management_db	1 row(s) affected	0.016 sec
✓ 2	19:19:39	USE orders_management_db	0 row(s) affected	0.000 sec
✓ 3	19:19:41	CREATE TABLE orders (ord_no INT PRIMARY KEY, purch_am...	0 row(s) affected	0.032 sec
✓ 4	19:19:47	INSERT INTO orders VALUES (70001,150.50,'2012-10-05',3005,500...	12 row(s) affected Records: 12 Duplicates: 0 Warnings: 0	0.015 sec

Program:

```
25      -- Step 3: WHERE Clause
26  ●    SELECT *
27      FROM orders
28      WHERE purch_amt > 2000;
```

Output:

Result Grid

Filter Rows:

Edit:

	ord_no	purch_amt	ord_date	customer_id	salesman_id
▶	70003	2480.40	2012-10-10	3009	5003
	70005	2400.60	2012-07-27	3007	5001
	70008	5760.00	2012-09-10	3002	5001
	70013	3045.60	2012-04-25	3002	5001
✱	NULL	NULL	NULL	NULL	NULL

5

19:20:45

SELECT * FROM orders WHERE purch_amt > 2000 LIMIT 0, 1000

4 row(s) returned

0.000 sec / 0.000 sec

Program:

```
29      -- Step 4: WHERE Clause with Date
30  ●    SELECT *
31      FROM orders
32      WHERE ord_date = '2012-09-10';
```

Output:

Result Grid

Filter Rows:

Edit:

	ord_no	purch_amt	ord_date	customer_id	salesman_id
▶	70007	948.50	2012-09-10	3005	5002
	70008	5760.00	2012-09-10	3002	5001
	70009	270.65	2012-09-10	3001	5005
●	NULL	NULL	NULL	NULL	NULL

6

19:21:31

SELECT * FROM orders WHERE ord_date = '2012-09-10' LIMIT 0, 1...

3 row(s) returned

0.000 sec / 0.000 sec

Program:

```
33      -- Step 5: WHERE Clause with Salesman
34      SELECT *
35      FROM orders
36      WHERE salesman_id = 5001;
```

Output:

Result Grid					
		Filter Rows:		Edit:	
	ord_no	purch_amt	ord_date	customer_id	salesman_id
▶	70002	65.26	2012-10-05	3002	5001
	70005	2400.60	2012-07-27	3007	5001
	70008	5760.00	2012-09-10	3002	5001
	70013	3045.60	2012-04-25	3002	5001
✱	NULL	NULL	NULL	NULL	NULL

7 19:22:20 SELECT * FROM orders WHERE salesman_id = 5001 LIMIT 0, 1000 4 row(s) returned

0.016 sec / 0.000 sec

Program:

```
37      -- Step 6: ORDER BY Purchase Amount
38      SELECT *
39      FROM orders
40      ORDER BY purch_amt DESC;
```

Output:

Result Grid					
		Filter Rows:		Edit:	
	ord_no	purch_amt	ord_date	customer_id	salesman_id
▶	70008	5760.00	2012-09-10	3002	5001
	70013	3045.60	2012-04-25	3002	5001
	70003	2480.40	2012-10-10	3009	5003
	70005	2400.60	2012-07-27	3007	5001
	70010	1983.43	2012-10-10	3004	5006
	70007	948.50	2012-09-10	3005	5002
	70009	270.65	2012-09-10	3001	5005
	70012	250.45	2012-06-27	3008	5002
	70001	150.50	2012-10-05	3005	5002
	70004	110.50	2012-08-17	3009	5003
	70011	75.29	2012-08-17	3003	5007
	70002	65.26	2012-10-05	3002	5001
✱	NULL	NULL	NULL	NULL	NULL

8 19:23:58 SELECT * FROM orders ORDER BY purch_amt DESC LIMIT 0, 10... 12 row(s) returned 0.000 sec / 0.000 sec

Program:

```
41 -- Step 7: ORDER BY Order Date
42 • SELECT *
43 FROM orders
44 ORDER BY ord_date ASC;
```

Output:

Result Grid					
Filter Rows:					
Edit:					
	ord_no	purch_amt	ord_date	customer_id	salesman_id
▶	70013	3045.60	2012-04-25	3002	5001
	70012	250.45	2012-06-27	3008	5002
	70005	2400.60	2012-07-27	3007	5001
	70004	110.50	2012-08-17	3009	5003
	70011	75.29	2012-08-17	3003	5007
	70007	948.50	2012-09-10	3005	5002
	70008	5760.00	2012-09-10	3002	5001
	70009	270.65	2012-09-10	3001	5005
	70001	150.50	2012-10-05	3005	5002
	70002	65.26	2012-10-05	3002	5001
	70003	2480.40	2012-10-10	3009	5003
	70010	1983.43	2012-10-10	3004	5006
•	NULL	NULL	NULL	NULL	NULL

9 19:24:57 SELECT * FROM orders ORDER BY ord_date ASC LIMIT 0, 1000 12 row(s) returned 0.000 sec / 0.000 sec

Program:

```
45 -- Step 8: SUM()
46 • SELECT SUM(purch_amt) AS total_revenue
47 FROM orders;
```

Output:

	total_revenue
▶	17541.18

10 19:25:45 SELECT SUM(purch_amt) AS total_revenue FROM orders LIMIT 0... 1 row(s) returned 0.000 sec / 0.000 sec

Program:

```
48 -- Step 9: AVG()
49 • SELECT AVG(purch_amt) AS average_order
50 FROM orders;
```

Output:

	average_order
▶	1461.765000

✓ 11 19:26:28 SELECT AVG(purch_amt) AS average_order FROM orders LIMIT ... 1 row(s) returned 0.000 sec / 0.000 sec

Program:

```
51      -- Step 10: MAX()
52 ●    SELECT MAX(purch_amt) AS highest_order
53      FROM orders;
```

Output:

	highest_order
▶	5760.00

✓ 12 19:27:21 SELECT MAX(purch_amt) AS highest_order FROM orders LIMIT 0,... 1 row(s) returned 0.015 sec / 0.000 sec

Program:

```
54      -- Step 11: MIN()
55 ●    SELECT MIN(purch_amt) AS lowest_order
56      FROM orders;
```

Output:

	lowest_order
▶	65.26

✓ 13 19:28:18 SELECT MIN(purch_amt) AS lowest_order FROM orders LIMIT 0, ... 1 row(s) returned 0.000 sec / 0.000 sec

Program:

```
57      -- Step 12: COUNT()
58 ●    SELECT COUNT(*) AS total_orders
59      FROM orders;
```

Output:

Result Grid	
	total_orders
▶	12

✓ 14 19:29:00 SELECT COUNT(*) AS total_orders FROM orders LIMIT 0, 1000 1 row(s) returned 0.016 sec / 0.000 sec

Program:

```

60      -- Step 13: GROUP BY Salesman
61      •   SELECT salesman_id,
62             SUM(purch_amt) AS total_sales
63      FROM orders
64      GROUP BY salesman_id;

```

Output:

Result Grid			Filter Rows:
	salesman_id	total_sales	
▶	5002	1349.45	
	5001	11271.46	
	5003	2590.90	
	5005	270.65	
	5006	1983.43	
	5007	75.29	

✓ 15 19:29:45 SELECT salesman_id, SUM(purch_amt) AS total_sales FROM ... 6 row(s) returned 0.000 sec / 0.000 sec

Program:

```

65      -- Step 14: GROUP BY Customer
66      •   SELECT customer_id,
67             SUM(purch_amt) AS total_purchase
68      FROM orders
69      GROUP BY customer_id;

```

Output:

	customer_id	total_purchase	
▶	3005	1099.00	
	3002	8870.86	
	3009	2590.90	
	3007	2400.60	
	3001	270.65	
	3004	1983.43	
	3003	75.29	
	3008	250.45	

✓ 16 19:30:35 SELECT customer_id, SUM(purch_amt) AS total_purchase FR... 8 row(s) returned 0.016 sec / 0.000 sec

Program:

```
70      -- Step 15: GROUP BY + MAX()
71  ●    SELECT customer_id,
72          MAX(purch_amt) AS highest_purchase
73  FROM orders
74  GROUP BY customer_id;
```

Output:

	customer_id	highest_purchase
▶	3005	948.50
	3002	5760.00
	3009	2480.40
	3007	2400.60
	3001	270.65
	3004	1983.43
	3003	75.29
	3008	250.45

✓ 17 19:32:42 SELECT customer_id, MAX(purch_amt) AS highest_purchase ... 8 row(s) returned 0.000 sec / 0.000 sec

Program:

```
75      -- Step 16: GROUP BY + HAVING
76  ●    SELECT salesman_id,
77          SUM(purch_amt) AS total_sales
78  FROM orders
79  GROUP BY salesman_id
80  HAVING SUM(purch_amt) > 3000;
```

Output:

	salesman_id	total_sales
▶	5001	11271.46

✓ 18 19:33:30 SELECT salesman_id, SUM(purch_amt) AS total_sales FROM ... 1 row(s) returned 0.000 sec / 0.000 sec

Program:

```

81      -- Step 17: GROUP BY + HAVING
82  ●    SELECT customer_id,
83          SUM(purch_amt) AS total_purchase
84      FROM orders
85      GROUP BY customer_id
86      HAVING SUM(purch_amt) > 2500;

```

Output:

	customer_id	total_purchase
▶	3002	8870.86
	3009	2590.90

✓ 19 19:34:18 SELECT customer_id, SUM(purch_amt) AS total_purchase FR... 2 row(s) returned

0.000 sec / 0.000 sec

Program:

```

87      -- Step 18: GROUP BY + HAVING
88  ●    SELECT customer_id,
89          COUNT(*) AS total_orders
90      FROM orders
91      GROUP BY customer_id
92      HAVING COUNT(*) > 1;

```

Output:

	customer_id	total_orders
▶	3005	2
	3002	3
	3009	2

✓ 20 19:36:29 SELECT customer_id, COUNT(*) AS total_orders FROM order... 3 row(s) returned

0.000 sec / 0.000 sec

Program:


```

93      -- Step 19: GROUP BY + HAVING + ORDER BY
94      SELECT customer_id,
95              SUM(purch_amt) AS total_purchase
96      FROM orders
97      GROUP BY customer_id
98      HAVING SUM(purch_amt) > 1000
99      ORDER BY total_purchase DESC;

```

Output:

Result Grid   Filter Rows:		
	customer_id	total_purchase
▶	3002	8870.86
	3009	2590.90
	3007	2400.60
	3004	1983.43
	3005	1099.00

21 19:37:55 SELECT customer_id, SUM(purch_amt) AS total_purchase FR... 5 row(s) returned 0.000 sec / 0.000 sec

Program:

```

100      -- Step 20: BETWEEN + HAVING
101      SELECT customer_id,
102              MAX(purch_amt) AS max_purchase
103      FROM orders
104      GROUP BY customer_id
105      HAVING MAX(purch_amt) BETWEEN 2000 AND 6000;

```

Output:

Result Grid   Filter Rows:		
	customer_id	max_purchase
▶	3002	5760.00
	3009	2480.40
	3007	2400.60

22 19:38:52 SELECT customer_id, MAX(purch_amt) AS max_purchase FR... 3 row(s) returned 0.000 sec / 0.000 sec

Program:

```
106      -- Step 21: COUNT + HAVING
107  ●    SELECT salesman_id,
108          COUNT(*) AS total_orders
109  FROM orders
110  GROUP BY salesman_id
111  HAVING COUNT(*) >= 2;
```

Output:

	salesman_id	total_orders
▶	5002	3
	5001	4
	5003	2

✓ 23 19:39:41 SELECT salesman_id, COUNT(*) AS total_orders FROM order... 3 row(s) returned

0.016 sec / 0.000 sec

Program:

```
112      -- Step 22: MAX + GROUP BY + HAVING
113  ●    SELECT ord_date,
114          MAX(purch_amt) AS highest_purchase
115  FROM orders
116  GROUP BY ord_date
117  HAVING MAX(purch_amt) > 2000;
```

Output:

	ord_date	highest_purchase
▶	2012-10-10	2480.40
	2012-07-27	2400.60
	2012-09-10	5760.00
	2012-04-25	3045.60

✓ 24 19:40:51 SELECT ord_date, MAX(purch_amt) AS highest_purchase FR... 4 row(s) returned

0.000 sec / 0.000 sec